

Week 1 Learning Report

1. Why ViT?

CNNs process an image through small local windows, so relationships between distant regions only appear in deep layers. The Vision Transformer (ViT) adapts the Transformer from language models to images. Every part of the image can interact with every other part from the first layer, so global structure is captured early. ViTs are also the backbone of self-supervised models such as DINOv2, which learn useful features without labels.

2. How image patchification works

A Transformer takes a sequence of tokens, so the image is first cut into small, non-overlapping square patches that are read in order (left to right, top to bottom).

- Number of patches = $(\text{image height} \div \text{patch size}) \times (\text{image width} \div \text{patch size})$.
- Each patch is flattened into a vector of $\text{patch height} \times \text{patch width} \times \text{channels}$ values.
- The image size must be a multiple of the patch size, otherwise the image is resized.
- The original ViT uses 16×16 patches. **DINOv2 uses 14×14 patches.**

3. Patch embeddings

Each flattened patch is passed through one shared linear layer that maps it to a vector of fixed length D , the model width ($D = 384$ for DINOv2 ViT-S/14). This vector is the patch embedding, or token.

- The same layer is applied to every patch.
- D stays constant through the whole network.
- At this stage a token describes only its own patch. Context is added by the Transformer layers.

4. Positional embeddings

Self-attention does not know the order of its inputs, so position information must be added. A learned position embedding of length D , one per token position, is added to each token. Each token then carries both its content and its location. ViT learns these as a 1D sequence and infers the 2D layout during training. If the input size differs from training, DINOv2 interpolates the position embeddings to fit.

5. CLS token

The CLS token is an extra learnable vector placed in front of the patch tokens. It does not come from the image, and it starts as the same vector for every image. Through attention in every layer it gathers information from all patches, so at the output it is a summary of the whole image. It is used as the global image representation, for example for classification and similarity search. There is one CLS token per image.

6. ViT Transformer architecture

ViT is an encoder-only Transformer: a stack of identical blocks (12 in ViT-S/14). Each block contains:

1. **Multi-head self-attention.** Each token produces a query, key and value. Attention weights come from comparing queries with keys, and each token's output is the weighted sum of the values. Multi-head means several attention operations run in parallel (6 in ViT-S/14), so different heads can capture different relationships.
2. **MLP (feed-forward network).** A two-layer network applied to each token separately.

Each part is preceded by LayerNorm and wrapped in a residual (skip) connection, which stabilizes training. The number of tokens and their dimension do not change, so the input and output of every block have the same shape.

7. CLS vs patch tokens

	CLS token	Patch tokens
Count	1 per image	One per patch
Describes	The whole image	One local region, with context from the rest of the image
Used for	Classification, similarity search, clustering	Localization, segmentation, part discovery
Limitation	No spatial detail	Many vectors per image

8. What DINOv2 is

DINOv2 is a family of Vision Transformers trained with self-supervised learning, meaning no labels or captions are used, to produce general-purpose image features. Training combines two ideas:

- **Self-distillation (DINO):** a student network is trained to match the output of a teacher network on different crops of the same image. The teacher is a slowly updated moving average of the student.
- **Masked patch prediction (iBOT):** some patches are hidden from the student, which must predict the teacher's representation for them, improving local features.

The model sizes are small, base, large and giant (widths 384, 768, 1024, 1536), all with patch size 14. This project uses **DINOv2 ViT-S/14** (12 layers, 6 heads, width 384) as a frozen feature extractor, so its weights are never updated.

9. What DINOv2 outputs

For one input image (B = batch size, N = number of patches, $D = 384$ for ViT-S/14), the model returns:

Output	Shape	Meaning
last_hidden_state	$(B, N + 1, D)$	All tokens after the final layer
CLS token	(B, D)	<code>last_hidden_state[:, 0]</code> , global representation
Patch tokens	(B, N, D)	<code>last_hidden_state[:, 1:]</code> , local representations
pooler_output	(B, D)	The CLS token as a separate output

Patch tokens follow reading order, so they can be reshaped into a grid of $(\text{image height} \div \text{patch size}) \times (\text{image width} \div \text{patch size})$. Embeddings are compared with cosine similarity, $\cos(x, y) = x \cdot y / (\|x\| \|y\|)$, which is 1 when two vectors point in the same direction.

10. How DINOv2 will be used in this project

The project uses DESI Legacy Imaging Surveys DR10 galaxy images (g, r, i, z bands) and Galaxy Zoo morphology labels. DINOv2 is used as a frozen feature extractor:

1. **Prepare:** build a 3-channel image from the bands, resize it and normalize it.
2. **Extract:** obtain the CLS token and patch tokens for each galaxy.
3. **Explore:** inspect the embeddings with PCA and nearest-neighbour search.
4. **Predict:** train simple models (k-NN, linear probe) on the frozen embeddings to predict Galaxy Zoo labels. The labels are used only for evaluation.
5. **Locate:** use patch tokens and attention maps to find structures such as arms, bars and companions.